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Subject: Urban Heat Mitigation Decision-Support Toolkit for Omaha
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Audience

Primary users are Omaha municipal planners, urban forestry staff, sustainability and climate resilience personnel, organizations involved in heat mitigation planning and investment, and GIS professionals supporting municipal planning. Secondary users include researchers and other GIS professionals.

Intention

Develop a configurable GIS-based decision-support toolkit that identifies areas in Omaha where different urban heat mitigation strategies may have the greatest potential benefit. The system will combine land surface temperature, tree canopy, impervious surface and land cover, population exposure, and local geographic context. It will explain why an area is prioritized, identify potential intervention categories, and show how priorities change when planning criteria are weighted differently. The toolkit will support planning decisions rather than produce definitive engineering or planning recommendations.

Geographic Extent

Omaha, Nebraska, using a defined study area established during data preparation. Local context may include applicable Douglas County or Omaha GIS data. The final analytical extent and spatial unit will be documented after data evaluation.

Deliverables

The final deliverable will be a locally executable, interactive web-based decision-support prototype supported by a reproducible geospatial data and analysis workflow.

Deliverable	Specification
Interactive decision-support web application	Interactive map for exploring heat conditions and priority areas, inspecting contributing criteria, and viewing potential intervention categories.
Configurable Multi-Criteria Analysis	Normalized criteria with adjustable weights and at least two planning scenarios that demonstrate how priorities change.
Spatial database	PostgreSQL/PostGIS database containing processed source data, derived spatial indicators, analytical criteria, and priority results.
Automated Python workflow	Scripts for data acquisition where practical, preprocessing, spatial metrics, normalization, analysis, and repeatable result generation.
Core spatial indicators	Heat intensity, tree canopy deficit, impervious surface, population exposure, land cover, and applicable site or intervention constraints.
Application summaries and export	Core summary statistics and charts, with export of analytical results where feasible after the core application is stable.
Project documentation	Data sources, methods, assumptions, architecture, limitations, setup instructions, and reproducibility documentation.
Source repository	Source code, configuration, documentation, and project history maintained in the GEOG_778 GitHub repository.

Budget

Planning assumption: approximately 160 hours of project effort. The budget is controlled by establishing a minimum viable product first and treating advanced capabilities as optional.

Work Area	Hours
Data acquisition and evaluation	15
Data preprocessing and ETL	25
PostgreSQL/PostGIS database development	15
Spatial metrics and Multi-Criteria Analysis	30
Web application and interactive mapping	35
Testing and quality assurance	15
Documentation and repository preparation	15
Final revisions and delivery	10
Total	**160**

Technology & Extraneous Costs

Category	Planned Technology or Resource	Additional Cost
Data	USGS Landsat-derived Land Surface Temperature; USDA Forest Service Tree Canopy Cover; MRLC Annual NLCD; U.S. Census ACS; Omaha/Douglas County GIS where applicable.	$0 planned
Data processing and analysis	Python; GeoPandas; Rasterio; Shapely; Pandas; NumPy; PyProj; additional Python packages as required.	$0 planned
Database	PostgreSQL and PostGIS.	$0 planned
Application	JavaScript; HTML/CSS; Kepler.gl or a comparable geospatial visualization library; Python API/application layer.	$0 planned
Development	Git; GitHub; Visual Studio Code; Python virtual environment and dependency management.	$0 planned
Hosting	Local development and execution environment for the core product. Public deployment is optional.	$0 planned
Training	No dedicated paid training is planned. Time for learning unfamiliar components is included within the 160-hour project budget.	$0 planned

No paid hosting, proprietary GIS software, paid data, or additional hardware is required for the core product, assuming existing computer and internet access. The project therefore has a planned additional monetary cost of $0.

Product Development & Delivery

Development will proceed from data evaluation and prototyping toward a working analytical workflow and then the interactive application. Source code, configuration, documentation, and project history will be maintained in the GEOG_778 GitHub repository.

- Data sources will be acquired and evaluated for availability, date, spatial scale, resolution, coverage, licensing, and suitability for the intended analysis.

- Python processing will standardize coordinate reference systems, spatial extent, resolution, attributes, and analytical formats before loading data into PostGIS.
- Spatial indicators will be generated from the processed datasets and normalized for use in Multi-Criteria Analysis.
- The analytical engine will support configurable criteria weights and scenario comparison. Where live recalculation is impractical, precomputed scenario results may be used.
- The web application will provide interactive map exploration, spatial inspection, criteria inspection, scenario controls, and core summary information.
- Testing and review will occur before final delivery, with revisions prioritized toward correctness, clarity, reproducibility, and completion of the minimum viable product.
- The core system will run locally. Public deployment may be considered only after the required analysis, application functionality, testing, and documentation are complete.

Architectural/Software Diagram

The architecture separates data management, analysis, and visualization so that the analytical workflow can be rerun with updated datasets or assumptions without rebuilding the entire visualization layer.

DATA SOURCES	PROCESSING	DATABASE	ANALYSIS	APPLICATION	USER
USGS Landsat LST USDA Tree Canopy Cover MRLC Annual NLCD U.S. Census ACS Omaha / Local GIS	Python ETL GeoPandas Rasterio Shapely Pandas / NumPy / PyProj	PostgreSQL PostGIS Processed and derived spatial data	Python Analysis Engine Spatial Metrics Normalization Multi-Criteria Analysis Scenario Analysis	Python API / Application Interface JavaScript Web Application HTML / CSS Kepler.gl or similar mapping library	Planner / Decision Maker Map selections Criteria inspection Scenario settings Results review

DATA SOURCES → PYTHON ETL → POSTGRESQL / POSTGIS → ANALYSIS ENGINE → API → JAVASCRIPT / KEPLER.GL → USER

User scenario settings and map selections may flow back through the application interface to retrieve or generate updated analytical results. The architecture includes the local execution environment as the planned hosting context. No separate design software is required for the core application.

Known Unknowns / Project Controls

- Final common analysis year and exact geographic study boundary.
- Availability, resolution, licensing, and coverage of local Omaha datasets.
- Exact spatial unit or analysis grid used to calculate and compare indicators.
- Normalization method and final criteria weighting structure.
- Specific intervention categories and the site constraints needed to support them.
- Final JavaScript framework and whether Kepler.gl provides all required interactions.
- Whether scenario changes require live recalculation or can use precomputed scenario results.

These items will be resolved during data evaluation and prototyping. If a capability threatens the core deliverable, it will be reduced or moved to optional scope rather than delaying the working prototype.

Timeline

The development timeline follows the course cadence from the Project Plan due September 22 through the Final Demonstration Video and Executive Report due December 1. Status reports, the project mockup/user testing milestone, and final demonstration provide checkpoints for scope, functionality, testing, and revision.

Week / Milestone	Focus	Planned Work
Week 1 **Sep 17 to Sep 22**	**Project plan and scope**	Finalize project aims, audience, geographic extent, deliverables, budget, technology, and scope controls. Confirm the minimum viable product and document initial data sources and risks. Submit the Project Plan September 22.
Week 2 **Sep 23 to Sep 29**	**Project pitch and technical setup**	Prepare the Elevator Pitch and refine the problem statement and intended user workflow. Establish GitHub structure, Python environment, and PostgreSQL/PostGIS database. Begin acquiring and evaluating priority datasets. Submit the Elevator Pitch September 29.
Week 3 **Sep 30 to Oct 6**	**Data evaluation and ETL prototype**	Evaluate data date, scale, resolution, coverage, licensing, and compatibility. Prototype CRS, extent, resolution, and attribute standardization. Load initial processed data into PostGIS. Document data limitations and report project progress in Status Report #1 October 6.
Week 4 **Oct 7 to Oct 13**	**Application mockup and user testing**	Develop the first application and map mockup. Establish the intended visual hierarchy, map interactions, criteria inspection workflow, and scenario controls. Conduct initial user testing and use findings to refine the application design. Complete Project Mockup/User Testing October 13.
Week 5 **Oct 14 to Oct 20**	**Spatial indicators and analysis foundation**	Finalize the analytical unit or grid. Develop spatial metrics for heat, canopy, impervious surface, population exposure, land cover, and applicable site characteristics. Establish and document the initial normalization approach. Begin implementing the analysis engine.
Week 6 **Oct 21 to Oct 27**	**Multi-Criteria Analysis**	Implement normalization and Multi-Criteria Analysis. Test initial weighting structures and generate the first priority results. Validate calculations and inspect results for data or methodological problems. Use findings to refine criteria and report progress in Status Report #2 October 27.
Week 7 **Oct 28 to Nov 3**	**Scenario analysis and application integration**	Implement scenario comparison and refine intervention categories and criteria. Connect analytical results to the application. Implement interactive mapping, spatial inspection, criteria inspection, and core scenario controls. Report progress in Status Report #3 November 3.
Week 8 **Nov 4 to Nov 10**	**Application refinement and visualization**	Refine interface layout, map symbology, explanatory content, summary statistics, and scenario comparison. Resolve integration issues between the analysis, database, API, and application layers. Add results export if the core application is stable.
Week 9 **Nov 11 to Nov 17**	**Testing and quality assurance**	Test data processing, spatial calculations, normalization, weighting behavior, scenario results, application interactions, and documentation. Conduct peer or instructor review where available. Record issues and prioritize corrections. Report progress in Status Report #4 November 17.
Week 10 **Nov 18 to Nov 24**	**Revision and final documentation**	Complete corrections from testing and review. Finalize workflow documentation, assumptions, limitations, architecture documentation, setup instructions, and repository organization. Confirm that all minimum viable product requirements are met.

Week 11 **Nov 25 to Dec 1**	**Final validation and delivery**	Perform final validation and reproducibility checks. Clean project files and repository structure. Verify that the core workflow can be rerun and that the application presents the intended results. Prepare the final demonstration video and executive report. Submit both December 1.

NOTE: Final project completion extends through December 8 for the Project Reflection/Instructor Conference and Project Abstract, and December 9 for the Digital Portfolio. These deliverables are treated as post-production documentation and reflection rather than additional core application development.

Minimum Viable Product

- Defined Omaha study area and documented data inventory.
- Working public-data acquisition and Python processing workflow.
- PostgreSQL/PostGIS database containing core processed datasets.
- At least one defensible priority analysis using normalized criteria and configurable weights.
- At least two planning scenarios demonstrating how priorities change.
- Interactive web map showing priority areas and contributing criteria.
- Documentation of methods, assumptions, limitations, setup, and reproducibility.

Optional Scope

Historical change analysis, additional intervention categories, advanced sensitivity analysis, extensive charting, live recalculation of complex scenarios, full public deployment, and additional local datasets will be pursued only after the minimum viable product is functional.

Success Criteria

The project will be considered complete when the core workflow can acquire and process the selected datasets, store and analyze the resulting spatial information, generate configurable priority results, present those results through an interactive map, and document the methods and limitations well enough for another GIS professional to understand and reproduce the workflow.